

The LULC model & dataset card database

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Where we are, and what was advised

Done so far

- Weeks 2 and 3: a 4-class base map (Alpha Earth, 10 m) plus a **living hierarchy**: split / add a class, retrain on the fly.
- Week 4: *designed* the **Model Card** and **Dataset Card** schema and a catalogue, on paper, no code.

What was advised this week

- **Convert the design into code**: a database that stores all the cards.
- A dataset needs a **type**: *training* (just polygons) vs *inference* (just the input features).
- A model card carries **both**: its training data and its inference feature source.
- Keep extent a simple **bounding box** for now; keep improving the front end.
- Try splitting trees into **tea** / **non-tea**.

From a design to a running database

The schema we designed last week is now code the tool writes to on every retrain:

- **Two card schemas:** extended with the new `type` field and the model's two dataset references, plus a validator.
- **A catalogue that is the database:** it validates, writes and indexes cards, mints them from live training runs, and answers “which models cover my area”.
- The catalogue is a **git repository**; publishing pushes the cards to GitHub.
- New API endpoints and a full-screen **Model Zoo** browser on the front end.

The pipeline: retrain $\rightarrow$ card $\rightarrow$ git

One left-to-right flow, all reusing what we already had:

- ➊ **Retrain a node:** the existing split / add trainer.
- ➋ **Mint cards:** lift the held-out report, the classes, and the marked polygons into a Model Card and its Dataset Cards.
- ➌ **Store them in the catalogue,** validated on write.
- ➍ **Publish = git push:** commit the cards and push to the shared GitHub repository.

The week-5 change: typed datasets + two references

- **Training dataset:** the *labels* a model learned from (`type: training`). Polygons or pixel tables, with classes; no embedding needed.
- **Inference dataset:** the *input features* a model runs on (`type: inference`). The Alpha Earth 64-d embedding; no labels.
- A **Model Card links to both:** `training.datasets` (what it was taught) and `inference.dataset` (what it consumes), so a model is portable and auditable.

The database is GitHub

- The catalogue is a **git working tree** of a shared zoo repository, the same model Hugging Face uses for its card hub.
- **Publish = commit + push.** A user trains a model, inspects its card, and pushes it to the zoo; versioning, sharing and history come for free.
- Only the **JSON cards** are committed; the trained model binaries stay local. A card carries the metrics and provenance that make it useful without the binary.
- Live now: cards push to a real GitHub repository.

Browse, and pick a model for your area

The catalogue answers queries

- List every card in the zoo.
- Filter to the models whose extent **covers your area**.
- Open any card in full.

Extent: show the real footprint

- For **polygon data**, the extent *is* the polygons: we draw the prominent ones on the map, not a country-sized box (everything here is India, so “India-wide” says nothing).
- Feature sources (Alpha Earth / Tessera) are genuinely India-wide, so those show a *label*. Querying “is there a model for my area” is still a plain bbox overlap.

Serving the map: Earth Engine tiles, not a PNG

- The base model is linear, so it replays **as band math inside Earth Engine**, classified at native 10 m on the server.
- Realistic mode is served as **EE map tiles** (an XYZ url), not a fixed PNG: crisp at *any* zoom, *any* area size, with nothing downloaded to us.
- This is the “tile URL” ask: the model is `expressible_as_bandmath`, the hook for pushing a model to EE and serving its output as tiles.

The full-screen Model Zoo

A dedicated browser, so the cards aren't crammed into the sidebar:

- **Tabs:** Models and Datasets; a responsive grid of uniform card tiles (class swatches and a published / local pill).
- **Detail pane:** the classes it produces (with the legend), a per-class metrics table, and the linked **training** and **inference** datasets as clickable chips.
- **Use a model:** *apply to map* makes a zoo model live (it registers on the tree and the map re-classifies); the base map card resets to the 4 classes. The zoo is usable, not just a catalogue.
- **Actions:** “only for current view” (the area filter), *Show on map* for the extent, and **Publish** (single card or all).

Describe it, prove it, map it

A model card isn't just numbers; the user annotates it (the “model cards in ML” ask):

- **Description, intended use, limitations:** what it's for and where it breaks.
- **Evidence:** how the classes were annotated (drone, field photos, ...), the “how do you know?” the brief asked for.
- **Map each class to a standard:** an optional crosswalk to ESA WorldCover / USDA, picked from a built-in list (no codes to memorize, and any subset is fine), so a localized class stays interoperable without the model count blowing up.

Two guards the brief asked for, surfaced as live feedback on the cards:

- **Spread (spatial diversity):** Shannon entropy of where a dataset's polygons fall, in $[0, 1]$; flags “100 polygons, all one district” before it skews a model. (Diversity, not raw volume, is what generalizes on unseen regions.)
- **Class balance:** the support ratio across classes, flagged balanced / mild / imbalanced; if it's skewed, the user can retrain with **undersampling** or **oversampling** instead of the default class-weighting.

Tea / non-tea: a separability check

- Tried the suggested **tea / non-tea** split of trees.
- Sampled Alpha Earth at 169 hand-marked polygons (105 tea / 64 non-tea), held out whole polygons, fit the same LinearSVC used everywhere else.

held-out accuracy				0.934
tea	P 0.938	R 0.961	F 0.950	
non-tea	P 0.925	R 0.884	F 0.904	

What's in the catalogue

Models available

- Base map: greenery, water, built-up, barren (pan-India).
- Greenery split: crops, trees, shrubs.
- Barren split: barren, mining.

- **11 cards** in all (3 model cards and 8 dataset cards), each **validated** against the schemas and pushed to the GitHub zoo.
- Every model card carries its training datasets, its inference feature source, and a bbox extent; held-out metrics live on each card.

Datasets available

- Training: expert polygon sets (crops, trees, mining), the master expert pixel table, ESA WorldCover slices, extra water.
- Inference: the shared Alpha Earth feature source.

Thank you